

Implementing business logic when using PHP

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In this section, we go through an example of implementing business logic using PHP. This example is a simple login program named 3tierapplication that needs a username and password .

Although authentication and authorisation should be decoupled from the business logic, it does have a cross-coupling effect – it can impact the detail in the ‘view’ (the HTML pages) when building menus and specific page options.

With authentication, we bring authorisation. We need both authentication and authorisation to ensure valid access to the data. From a software architect perspective, the authentication and authorisation can be abstracted away entirely from the business logic, but this only makes sense for larger systems. For smaller systems, it is fine to integrate them as part of the business logic.

Four files have been used in this example to illustrate how the three layers are used in a PHP application.

File name	Description	Layer
File 1: login.html	Used to receive the username and password from the user.	Presentation
File 2: style.css	Keeps the informational content of a document separate from the details of how to display it.	Presentation

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File 3a: config.php	Configures the database with hostname and root details.	Database
File 3b: database	Contains a database named topic9 with a table called login. The login table includes username and password fields (VARCHAR(20)).	Database
File 4: authentication.php	Authenticates the received username and password from the index.html form with the saved credentials in the database. Displays 'Login successful' or 'Login failed' based on validation.	Business logic

Presentation layer

This login form is created to interact with the user. It will receive the username and password from the user and also contains empty field validation, which is written in JavaScript. The code for the login.html file (file 1) is given below:



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```
1  <html>
2  <head>
3      <title>PHP login system</title>
4
5      <link rel = "stylesheet" type = "text/css" href = "style.css">
6  </head>
7  <body>
8      <div id = "frm">
9          <h1>Login</h1>
10         <form name="f1" action = "authentication.php" onsubmit = "return validation()"
11             <p>
12                 <label> UserName: </label>
13                 <input type = "text" id ="user" name  = "user" />
14             </p>
15             <p>
16                 <label> Password: </label>
17                 <input type = "password" id ="pass" name  = "pass" />
18             </p>
19             <p>
20                 <input type = "submit" id = "btn" value = "Login" />
21             </p>
22         </form>
23     </div>
24     <!-- validation for empty field -->
25     <script>
26         function validation()
27         {
```



The code below is the **style.css** (file 2), which provides a more pleasant appearance to the login form with user-defined foreground colour and background colour. The CSS code for the style.css file is given below:



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```
1  body{
2      background: #eee;
3  }
4  #frm{
5      border: solid gray 1px;
6      width:25%;
7      border-radius: 2px;
8      margin: 120px auto;
9      background: white;
10     padding: 50px;
11 }
12 #btn{
13     color: #fff;
14     background: #337ab7;
15     padding: 7px;
16     margin-left: 70%;
17 }
```

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Business logic layer

The authentication.php file (file 3) handles the login form data that was sent through the index.html file. It validates the data sent through the login form. If the username and password match with the database, then the login will be successful; otherwise, the login will fail. The core functionality of the program – that is, authentication – is kept separated from the database layer and presentation layer. The code for the authentication.php file is below:



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```
1  <?php
2  include("config.php");
3
4  if(isset($_POST['sub']))
5  {
6      $uname = $_POST['uname'];
7      $upassword = $_POST['upassword'];
8
9      $res = mysqli_query($mysqli,"select* from login where uname='$username' and upassword='$password'");
10     $result=mysqli_fetch_array($res);
11     if($result)
12     {
13         echo "You have logged in Successfully ";
14
15
16     }
17     else
18     {
19         echo "failed ";
20     }
21 }
22 ?>
```

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Data layer

The following files that are related to the data layer are **config.php** (file 4) and a physical database created using phpMyAdmin (part of the WAMP/MAMP/LAMP server technology stack).

- phpMyAdmin was used to create a database and database table. The database name is '**topic9**'. The table name is '**login**'.
- config.php is the connection file that contains the connection code for database connectivity.

The code below is from the config.php file:
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NOTE: We will explore the database side of PHP in the next topic. This includes details regarding the username/password for the database.



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```
1  <?php
2
3  $databaseHost = 'localhost';//or localhost
4  $databaseName = topic9; // your db_name
5  $databaseUsername = 'root'; // root by default for localhost
6  $databasePassword = ''; // by default empty for localhost
7
8  $mysqli = mysqli_connect($databaseHost, $databaseUsername, $databasePassword, $databaseName);
9
10 ?>
```

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In the example 3tierapplication login program, which layer or layers would you find the following PHP code in?

```
<?php

$databaseHost = 'localhost';//or localhost
$databaseName = topic9; // your db_name
$databaseUsername = 'root'; // root by default for localhost
$databasePassword = ''; // by default empty for localhost

$mysqli = mysqli_connect($databaseHost, $databaseUsername, $databasePassword, $databaseName);

?>
```

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☒ The advantages of separate application layers

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